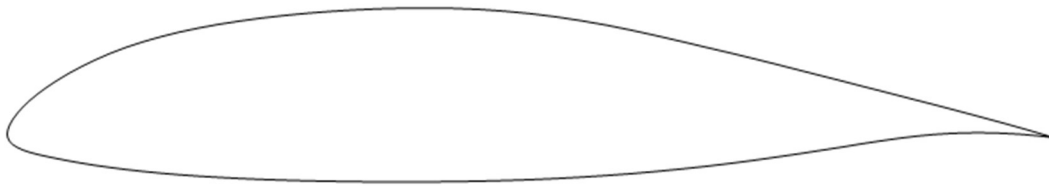


PART 1: THE CONSTANT STRENGTH VORTEX METHOD APPLIED TO SIMPLE AND TWO-ELEMENT AIRFOILS

This part focuses on the two-dimensional incompressible potential flow around a HQ 300 airfoil and a system consisting of two NACA 0015 airfoils, their geometries being given by the coordinates of discrete points listed in the txt files provided (for ever increasing number of divisions in powers of 2).

For all the questions below, the results obtained must be adequately commented and justified.

Consider initially the HQ 300 airfoil.



The following is requested:

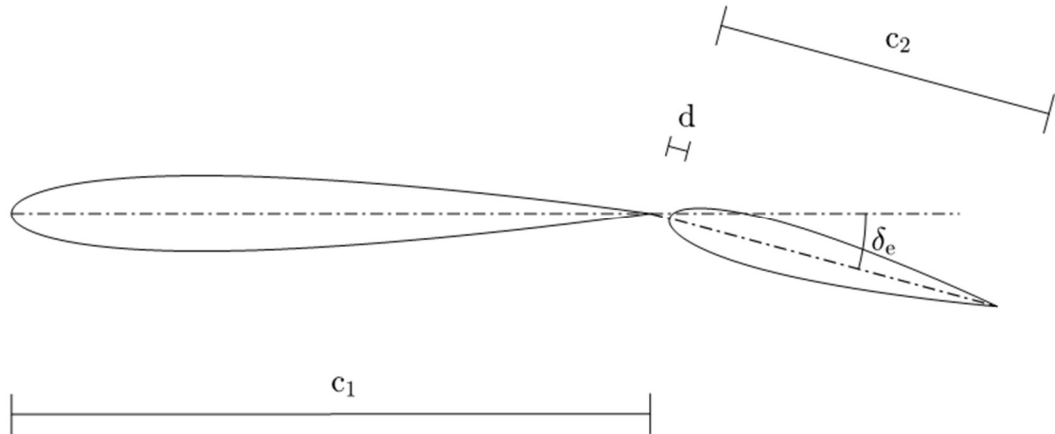
1. Obtain the lift coefficient C_l and the pitching moment coefficient about the quarter chord point $C_{m_{1/4}}$ for the angles of attack varying in a range $0 \leq \alpha \leq 8^\circ$.

HQ 300	$\alpha = 0^\circ$	$\alpha = 2^\circ$	$\alpha = 4^\circ$	$\alpha = 6^\circ$	$\alpha = 8^\circ$
C_l					
$C_{m_{1/4}}$					

Determine also the position of the aerodynamic center.

2. Using Kármán-Tsien's rule, determine the critical Mach number for $\alpha = 0^\circ, 2^\circ, 4^\circ$.
3. Compute the lift coefficient for $\alpha = 4^\circ$, for the following free stream Mach numbers:
 $M_\infty = M_{cr} - 0.15$, $M_\infty = M_{cr} - 0.10$, $M_\infty = M_{cr} - 0.05$, $M_\infty = M_{cr}$.

Consider now a system consisting of two symmetric NACA 0012 airfoils arranged in tandem as depicted in the figure below, with their respective chords $c_1 = 0.64$ and $c_2 = 0.34$, and gap $d = 0.02$. The angle δ_e can vary both up and down.



The following is requested:

- Obtain the lift coefficient C_l , and the pitching moment coefficient about the quarter chord point $C_{m_{1/4}}$, for zero elevator deflection, and the angle of attack varying in a range $0 \leq \alpha \leq 8^\circ$.

Compound Airfoil	$\alpha = 0^\circ$	$\alpha = 2^\circ$	$\alpha = 4^\circ$	$\alpha = 6^\circ$	$\alpha = 8^\circ$
C_l					
$C_{m_{1/4}}$					

- Compute the lift coefficient and the moment coefficient about the original quarter chord point, for an angle of attack $\alpha = 4^\circ$, and the elevator deflection angle varying in a range $0 \leq \delta_e \leq 16^\circ$.

Compound Airfoil	$\delta_e = 0^\circ$	$\delta_e = 4^\circ$	$\delta_e = 8^\circ$	$\delta_e = 12^\circ$	$\delta_e = 16^\circ$
C_l					
$C_{m_{1/4}}$					